

Pedro Fontanarro

Multidisciplinary Researcher & Software Engineer in Synthetic Biology | pfontanarro@gmail.com | +44 756 459 4770 | London, GB | fontanpink.github.io/Resume-PedroFontanarro/

Summary

Broadly skilled researcher with a strong foundation in computational/software engineering and biological sciences. Extensive experience developing Genetic Design Automation (GDA) tools and mathematical models for genetic circuit design acquired during Master's and PhD research. Currently expanding into advanced machine learning algorithm development at UCL—designing custom Gaussian process regression kernels, applying Bayesian inference with VAR regression for time-series analysis, and developing physics-informed neural networks to infer biological system dynamics. Excels in remote, multidisciplinary collaborations.

Skills

- **Software Engineering:** Python, Java, C++, JavaScript, R, Git, GitHub, GitLab, CI/CD, Docker, Kubernetes, Google Cloud, TensorFlow, PyTorch, scikit-learn
- **Optimization & Operations Research:** CPLEX, Gurobi, Pyomo, Linear Programming, Nonlinear Programming, Integer Programming
- **Web Scraping:** Scrapy, Splash, Selenium
- **Probabilistic Modeling:** Gaussian Process Regression, Bayesian Machine Learning
- **Data Science & Machine Learning:** Artificial Intelligence, Data Analysis, Neural Networks, Physics-Informed Neural Networks, Bayesian Inference, VAR Regression
- **Web Development:** HTML, Hugo, Jekyll
- **Databases:** SQL
- **High Performance Computing:** HPC, Parallel Computing, Cluster Computing, MPI, Job Scheduling, Distributed Computing

Work Experience

Computational Systems and Synthetic Biology Lab, University College London

London, UK

Postdoctoral Researcher

Jun 2023 — Present

- Developed robust Python packages for genetic design automation and modeling inference
- Created and adapted CI/CD pipelines for scalable computational biology workflows
- Applied and advanced machine learning (GP, VAR, NN) and Bayesian optimization algorithms to enhance predictive modeling and system design

Genetic Logic Lab, University of Boulder

Boulder, CO, USA

Postdoctoral Researcher

Aug 2022 — Jun 2023

- Enhanced iBioSim functionalities for genetic circuit modeling
- Developed robust mathematical models for genetic design
- Collaborated with international research teams using cloud-based tools

University of Utah

Salt Lake City, UT, USA

Research Assistant (PhD Researcher)

Aug 2019 — Aug 2022

- Simulated genetic regulatory networks using stochastic modeling, forecasting gene expression levels with 95% accuracy and guiding subsequent experimental validation by other team members.
- Guided experimental research through simulation-based predictions
- Collaborated on interdisciplinary synthetic biology projects

COMBINE Standards

Online

SBOL Editor

Jan 2019 — Jan 2022

- Led weekly meetings to coordinate SBOL standard updates
- Facilitated community-driven revisions and improvements to SBOL
- Enhanced interoperability in synthetic biology data exchange

University of Utah

Salt Lake City, UT, USA

Research Assistant (Master's Researcher)

Aug 2017 — Aug 2019

- Contributed to preliminary genetic circuit modeling tools
- Supported modeling and simulation of genetic regulatory systems
- Collaborated on interdisciplinary research initiatives

Northlands School

Buenos Aires, Argentina

Highschool Chemistry Teacher

Jan 2015 — Jan 2017

- Developed lesson plans that enhanced student engagement
- Mentored students in chemical research projects

Tarbut School

Buenos Aires, Argentina

Science and Mathematics Teacher

Jan 2014 — Jan 2015

- Designed and implemented creative educational programs
- Coordinated school-wide science and math competitions

Evolutionary Studies Laboratory, University of Buenos Aires

Buenos Aires, Argentina

Research Assistant

Jan 2010 — Jan 2014

- Coordinated field expeditions and permit negotiations
- Managed statistical programs and databases
- Trained new laboratory members

Volunteering

Data-Centric Biological Design & Engineering Interest Group

Oct 2024 — May 2025

Organizer

Organized a monthly seminar series under the Alan Turing Institute's banner to leverage AI in advancing biological system engineering. Unites experts from computer science, biology, and engineering to address global challenges in sustainable manufacturing, healthcare innovation, and environmental impact.

Biohacking BA

Jan 2013 — Jan 2017

Volunteer Organizer

Organized talks, workshops, hackathons, and DIY projects to promote innovation in science, engineering, and synthetic biology.

- Organized hackathons for SBOL and FAIR data practices
- Coordinated interdisciplinary teams and managed remote collaboration

University of Buenos Aires Biology Week

Jan 2010 — Jan 2012

Event Organizer

Coordinated the annual Biology Week to promote science careers among high school students.

Education

University of Utah

Ph.D.

- 3.8/4.0

Aug 2019 — Aug 2022

University of Utah

Master's

- 3.67/4.0

Aug 2017 — Aug 2019

University of Buenos Aires

Licentiate

- 8.39/10.00

Jan 2007 — Jan 2014

Profiles

GitHub

fontanapink

github.com/Fontanapink

ORCID

0000-0002-0535-2684

orcid.org/0000-0002-0535-2684

Google Scholar

Pedro Fontanarrosa

scholar.google.com/citations?hl=en&user=UemPJnYAAAAJ

Research Gate

Pedro Fontanarrosa

www.researchgate.net/profile/Pedro_Fontanarrosa3?ev=hdr_xprf

LinkedIn

Pedro Fontanarrosa

www.linkedin.com/in/pedro-fontanarrosa-37372474/

Certifications

- **Data Science Bootcamp** : THE ERDŐS INSTITUTE
- **Machine Learning A-Z™: AI, Python & R + ChatGPT Bonus [2023]** : Udemy
- **Optimization with Python: Solve Operations Research Problems** : Udemy
- **Optimization with Python: Complete Pyomo Bootcamp A-Z** : Udemy
- **AI and Meta-Heuristics (Combinatorial Optimization) Python** : Udemy
- **Modern Web Scraping with Python using Scrapy Splash Selenium** : Udemy
- **Deployment of Machine Learning Models** : Udemy
- **Pyomo Bootcamp: Python Optimization from Beginner to Advance** : Udemy
- **Theory of Gaussian Process Regression for Machine Learning** : Udemy

Projects

Synergistic Discovery and Design (SD2)

Genetic circuit design for extreme environments enabled by models extracted from petabyte+ perturbation analyses.

Jan 2018 — Jun 2022

iBioSim Development

Worked on iBioSim—a CAD tool for modeling, analysis, and design of genetic circuits supporting SBML and SBOL, including capabilities for multicellular and spatial models.

Advanced Machine Learning for Biological Systems

Developing and applying novel machine learning algorithms to model and predict the structure and dynamics of biological systems. Techniques include custom Gaussian process regression with covariance (kernel) designs, Bayesian inference with VAR regression for time-series analysis, and physics-informed neural networks for scalable inference with limited samples. Originally applied to microbial systems, these methods are generalizable to any biological system.

Detecting Fake News

A machine learning project that uses advanced algorithms to detect fake news. Developed in Python and hosted on GitHub.

Jan 2022 — Present

[github.com/Fontanapink/Detecting-Fake-News-A-Machine-Learning-Approach]

Awards & Recognitions

- **Fulbright and Argentine Presidential Fellowship in Science & Technology**: Awarded to pursue a master's degree in the United States starting Fall 2017.
- **Research and Communication Excellency Award**: Recognized for excellence in research and communication under the 'Beca Estímulo' scholarship.
- **Beca Estímulo (Encouragement Scholarship)**: Supported research and development tasks in genetics and ecology.